

The **IF function** is one of the most popular and useful logical functions in Excel, used to return one value if a condition is true and another value if it is false.

1. Core Syntax

The function operates based on a specific logic flow:

`=IF(logical_test, value_if_true, value_if_false)`

- **logical_test**: The condition you want to check (e.g., is a score greater than 35?).
 - **value_if_true**: The result returned if the test is met.
 - **value_if_false**: The result returned if the test is not met.
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2. Logical Operators Used in IF

To build the "logical_test," Excel uses standard comparison symbols:

- `=`: Equal to.
 - `>`: Greater than.
 - `<`: Less than.
 - `>=`: Greater than or equal to.
 - `<=`: Less than or equal to.
 - `<>`: Not equal to.
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3. Advanced Logical Techniques

Excel allows for more complex logical structures by combining the IF function with other tools:

- **Nesting**: You can place an IF function inside another (Nested IF) to test for multiple conditions.
 - **Combining with AND()**: You can use the `AND()` function inside an IF statement to ensure multiple criteria are all true before returning the "true" result.
 - **IFERROR()**: This is a specialized logical function used to catch and handle errors, returning a custom message instead of an error code like `#VALUE!`.
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4. Practical Example

In a student grading sheet, you might use the following logic to determine if a student passed:

- **Formula**: `=IF(G2>=35, "Pass", "Fail")`.

- **Execution:** If the value in cell **G2** is 35 or higher, the cell displays "Pass"; otherwise, it displays "Fail".

Related Conditional Functions

Beyond the basic IF function, Excel provides mathematical versions for specific data management tasks:

- **COUNTIF():** Counts the number of cells that meet a specific criterion.
- **SUMIF():** Adds the values in a range that meet a specific criterion.